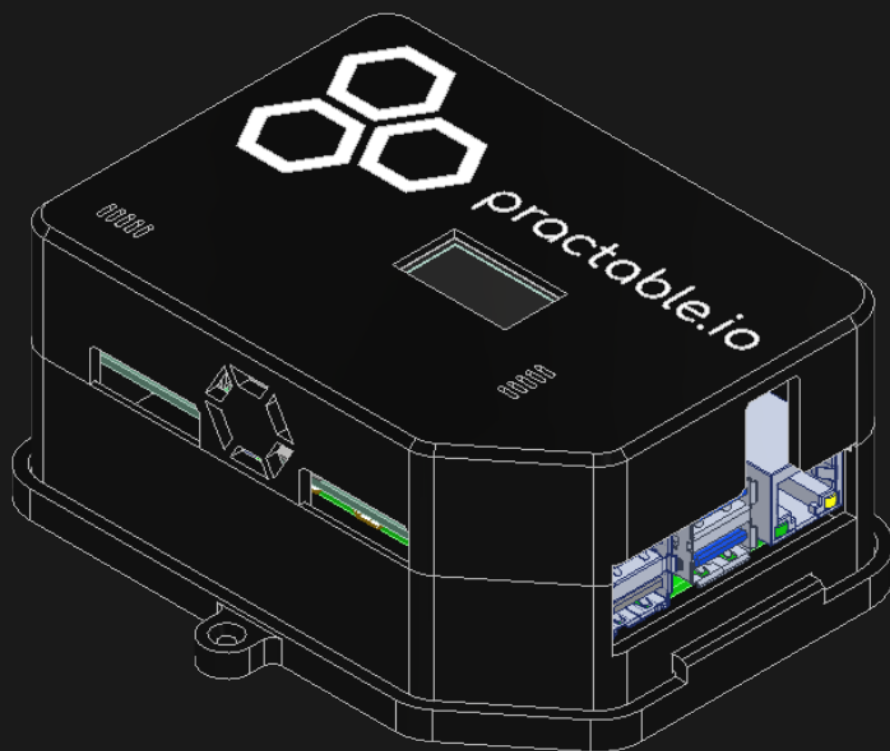


Practable: Portal

Instant IoT Connectivity for Remote Lab Experiments

User Guide



Version: 1.0.0

Portal Version: 1.2.0

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Research Use Notice

This equipment is supplied as prototype research hardware and is not classified as consumer electronic equipment. The intended use case may require modification or integration of the hardware to meet specific customer or application requirements.

1 Practable Portal: Introduction

Practable Portal is a turnkey solution for participating in the Practable.io Remote Labs project. It is pre-configured to stream a webcam feed and relay USB serial communication between a connected microcontroller and the Practable.io servers, reducing the complexity of deploying experiments as remote labs.

Designed for prototyping your own remote lab, Practable Portal includes an adjustable power supply designed to support higher-current loads such as stepper motors and servos, providing a compact and integrated solution for deploying teaching experiments using microcontroller-based systems to the browser based Practable.io Platform.

1.1 Features

- Fuse protected 24 V input with polarity protection
- Fuse protected 5 V output for microcontrollers and logic-level devices
- Fuse Protected User-adjustable output from 3.3 V to 24 V
- Integrated power supply for Raspberry Pi SBC
- Voltage and current monitoring
- Network connection status display
- Remote power cycle and reset functionality

1.2 System Components

- *Raspberry Pi* single-board computer (SBC)
- Pre-configured *Raspberry Pi* microSD card w/ *Raspberry Pi OS*
- Practable Portal Raspberry Pi HAT and power supply
- Practable Portal enclosure and mounting plate
- 24 V DC power supply

2 System Specification

2.1 Electrical Specification

IO	Symbol	Condition	Min	Typ.	Max	Units	Comments
<i>Supply</i>							
Supply Voltage	V_{in}		6	24	24	V	
Supply Current	I_{in}	@ $V_{in} = 24\text{ V}$	0.190		3.5	A	Fused @ 3.5 A
<i>Power Outputs</i>							
Unregulated V_{out}	V_{dd}				V_{in}	V	
SBC Power V_{out}	V_{sbc}	$I_{max} = 3.0\text{ A}$	4.9	5.0	5.1	V	Power to SBC
Experiment Power V_{out}	V_{exp}	$I_{max} = 3.0\text{ A}$	3.3	12	24	V	Set via Trimpot
<i>Digital IOs</i>							
Shutdown Send	DO_{ss}		0		3.3	V	Send shutdown/reboot signal to remote
Remote Reboot Input	DI_{rr}		0		3.3	V	Receive remote shutdown/reboot signal from remote

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3 Practable Portal: Hardware Definitions

3.1 IO Definition

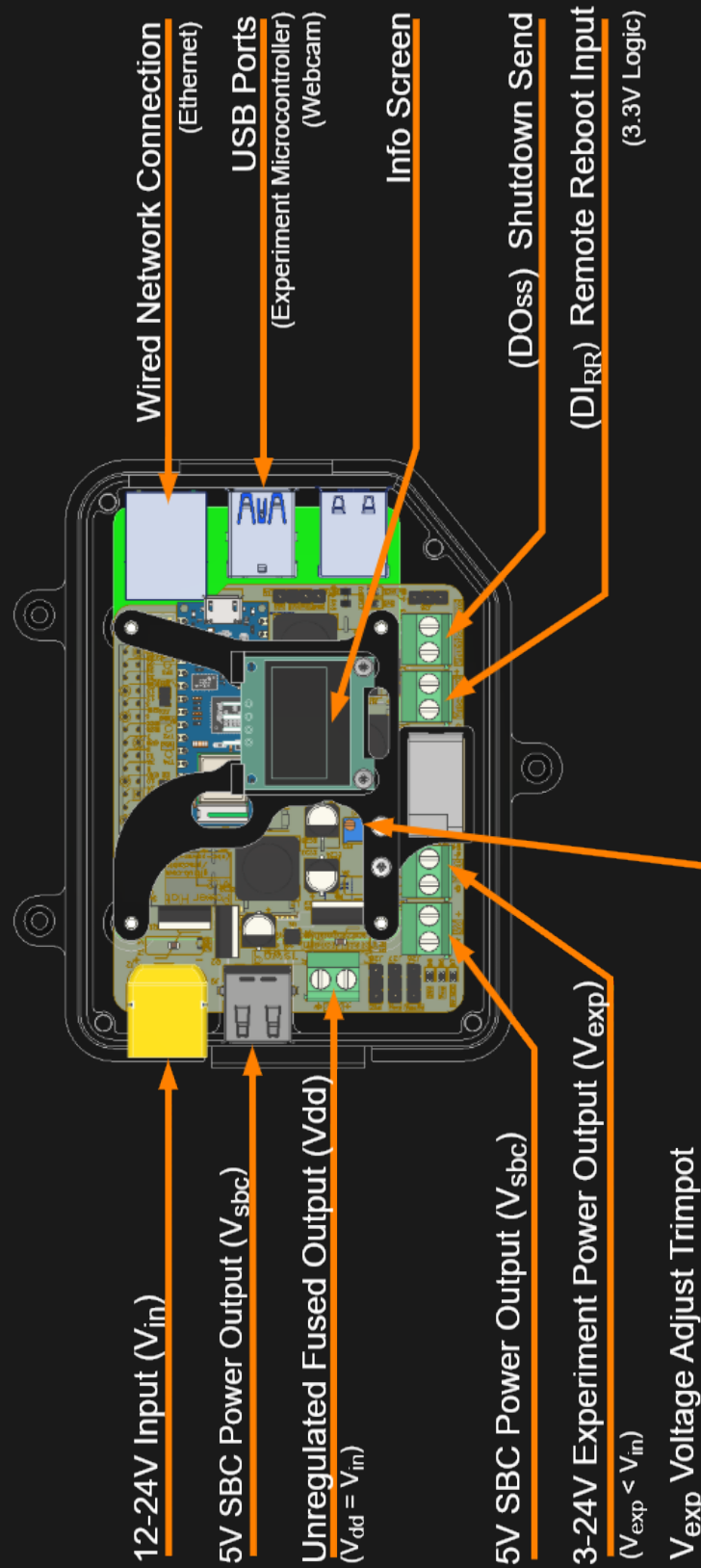


Figure 1: Portal IO Definitions

3.2 Hardware Description

3.2.1 Power Supply

The Practable Portal provides independent power outputs to help power your projects.

3.2.2 Adjustable Power Output V_{exp}

This high-current output can supply up to 3 A and is adjustable from 3.3 V up to the input supply voltage. The output voltage is adjusted using the *Voltage Adjust Trimpot* and a small flat-head screwdriver. Refer to *Section 3.3.2* for the voltage adjustment procedure.

3.2.3 5V SBC Power Output V_{sbc}

This fixed-voltage output powers the single-board computer (SBC) that runs the Portal software. It may also be used to power other external devices if required.

The output can supply up to 3 A total of which the SBC may draw up to 2.5 A.

3.3 Info Screen

The Info screen provides basic system information to assist with experiment setup, network monitoring, and troubleshooting potential system faults.

Three main screens are displayed in a continuous cycle while the equipment is powered on.

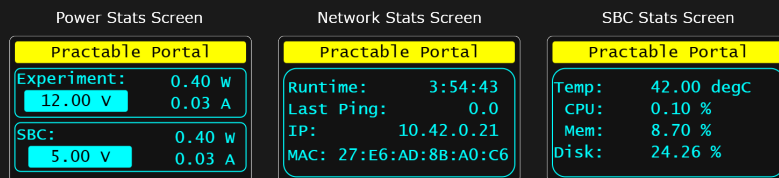


Figure 2: Info Screen Pages

3.3.1 Power Stats Screen

Displays status information for each power rail, including the adjustable experiment supply V_{exp} and the fixed 5 V SBC supply V_{sbc} for the Portal computer.

The displayed values are intended for basic power monitoring and fault finding only and should not be used for experimental data acquisition purposes.

3.3.2 Network Stats Screen

Displays network status information for the SBC, including *System Runtime*, *Assigned IP* and unit *MAC address*. Displays *time since last ping* to indicate valid internet connection.

An IP address will be displayed when the Practable Portal has established a valid network connection. This indicates that the Portal is connected to the local network and is able to communicate with Practable servers, provided a working internet connection is available.

3.3.3 SBC Stats Screen

Displays additional SBC status information, including *processor temperature*, *CPU*, *memory*, and *disk utilisation* statistics.

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4 Basic Setup

Practable Portal can be online in just a few simple steps.

4.1 Initial Setup

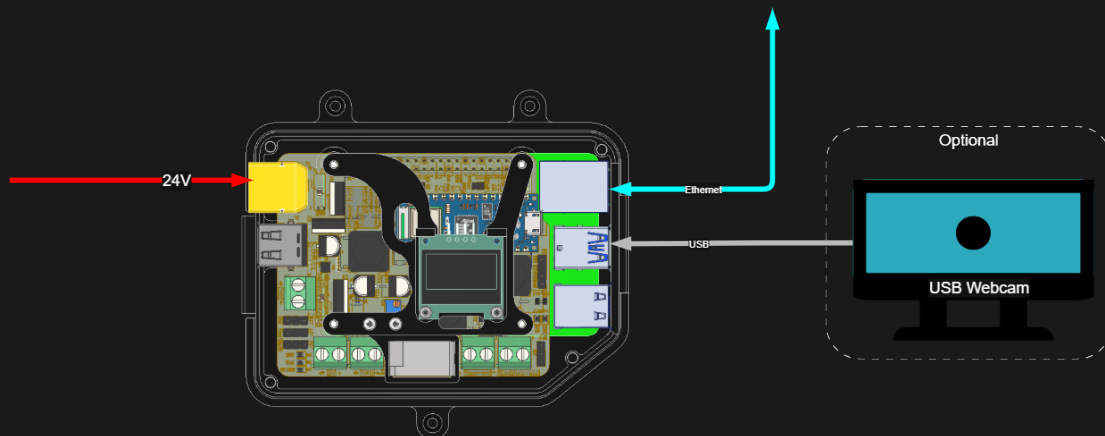


Figure 3: Portal Initial Setup

4.1.1 Network Connection

Connect an Ethernet cable from a LAN port to the Ethernet port on the Practable Portal. Within institutional or managed networks, permission from the network administrator may be required before connecting third-party devices to the wired network.

If the network administrator requires the MAC address of the Portal unit, power the device as described below and refer to the Network Stats page on the Info screen.

4.1.2 Webcam Connection (Optional)

If required, connect a USB webcam to the Practable Portal.

4.1.3 Power Connection

Connect the yellow XT60 connector on the Practable Portal chassis to the DC supply voltage. The input supply may range from 6 V to 24 V, with a nominal operating voltage of 24 V.

The input supply voltage determines the maximum available voltage of the adjustable experiment power output, V_{exp} .

4.2 Checking Operation

If the Practable Portal has obtained a valid IP address through the LAN connection, this will be displayed on the Network Stats page of the Info screen.

If a USB webcam has been connected, it should now be active and streaming video, ready to be accessed through the *Practable.io* web service.

5 Setting up an Experiment

Setting up an experiment with the Practable Portal can be as simple as connecting a microcontroller to one of the USB ports on the right-hand side of the unit.

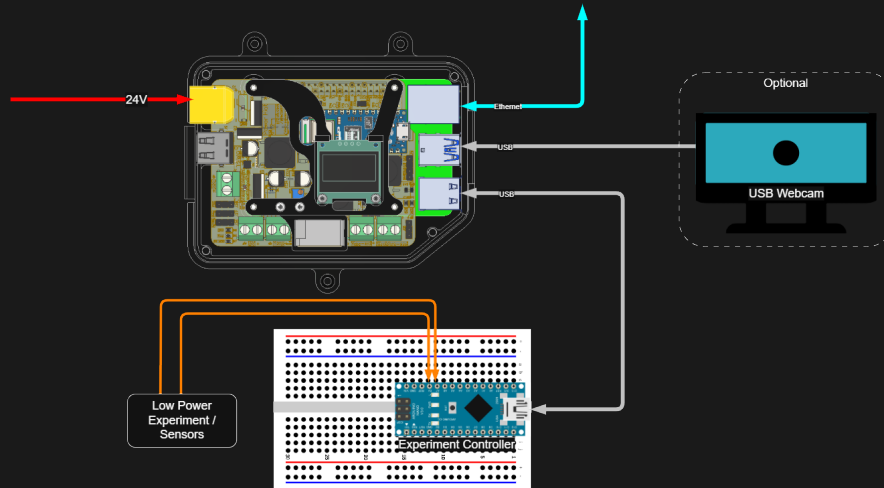


Figure 4: Portal Basic Experiment

If the Practable Portal is already powered when a microcontroller is connected, the unit may need to be power-cycled before it can begin streaming and accepting data over USB Serial.

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5.1 Using the Power Supply V_{exp}

For experiments requiring higher voltage or current than can be supplied via USB power alone, an additional power supply is available via the screw terminals on the Practable Portal.

This power supply should be kept electrically separate from the microcontroller's digital ground. This helps ensure that high return currents from devices such as motors do not interfere with microcontroller operation.

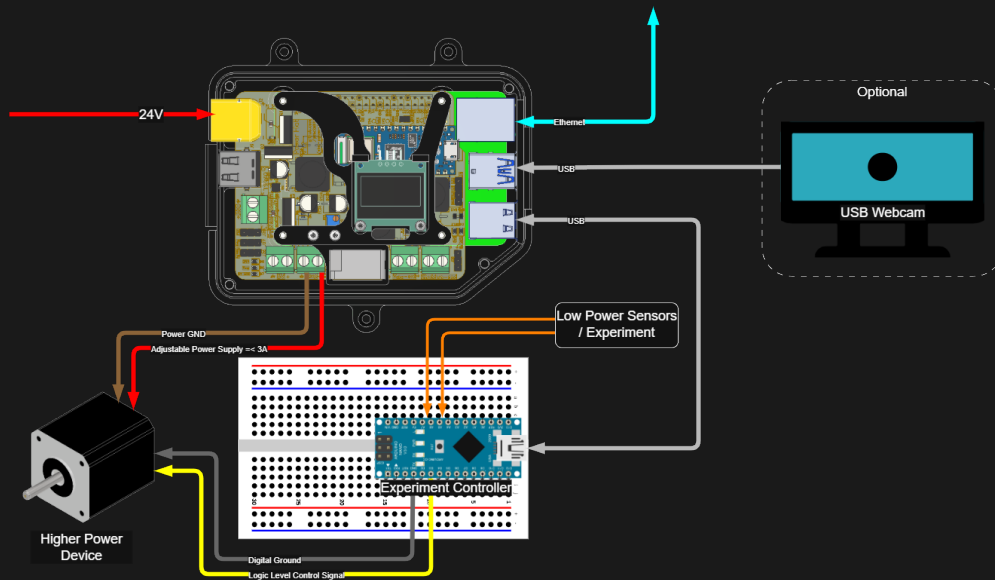


Figure 5: Using Portal Power Supply

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5.1.1 Adjusting the Output Voltage V_{exp}

The voltage supplied by the Experiment Power Rail (V_{exp}) can be adjusted using a flat-blade screwdriver to turn the *Voltage Adjust Trimpot* (pink circle).

The output voltage is shown on the display. Adjust the trimpot slowly to ensure the displayed value remains accurate and stable.

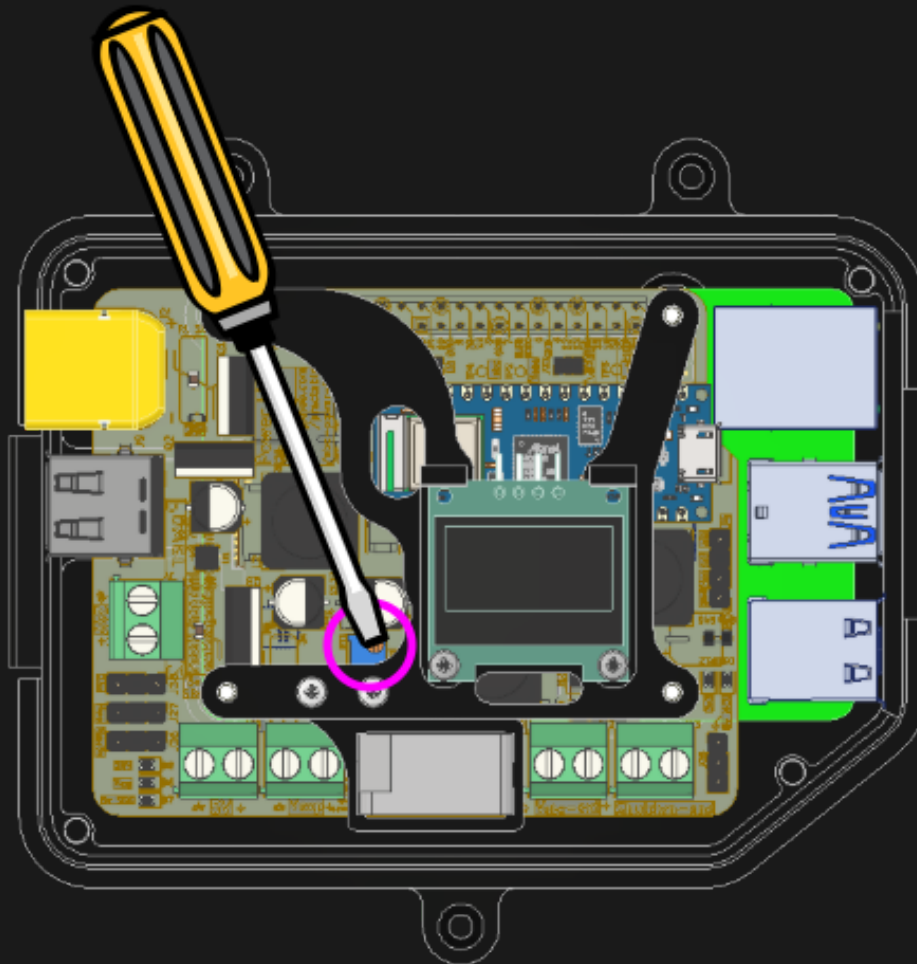


Figure 6: Adjusting the Voltage

If precise voltage setting is required, use an external multimeter to confirm correct voltage has been achieved.

6 Fault Finding & Debugging

6.1 No Network Connection

If Practable Portal does not have a valid network connection, it will not be assigned an IP address.

This will be indicated by a blue box in the IP field of the *Network Stats* screen, and will likely be accompanied by *Last Ping* showing any number greater than 0.0, or "None" if Portal has never had a valid network connection.

Network Stats Screen

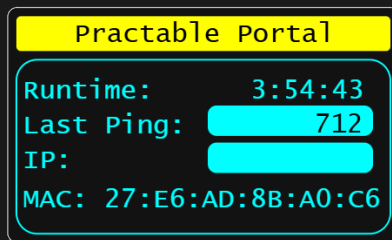


Figure 7: Missing LAN Screen

6.1.1 Remediation

Please contact your Network Administrator and ensure that the device is allowed to connect to the LAN. Network Administrators may need the unit's MAC address to ensure the Portal is permitted external internet access.

6.2 Network Connection, No Internet

Practable Portal may have a valid LAN connection, and have been assigned an IP address, but is still unable to connect to the internet. In this case, the IP address will be displayed, but *Last Ping* will show a number greater than 0.0 (Or "None") and will be highlighted.

Network Stats Screen

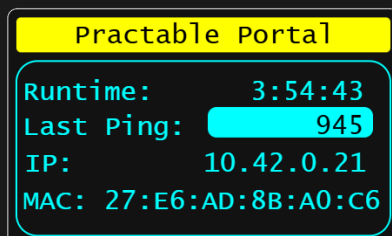


Figure 8: No Internet Screen

6.2.1 Remediation

- Please contact your Network Administrator and ensure that the device is placed on any whitelists required to enable external internet access.
- Check for valid internet access on other devices connected to the same network.
- Attempt to connect device using shared internet from another online device

6.3 Missing Data on Info Screen

If the power supply is unable to communicate with the Portal SBC for any reason, the Network Stats and SBC Stats screen will not populate with data, as shown below.

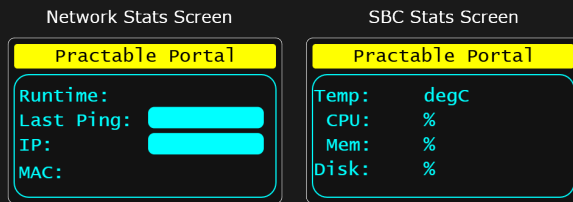


Figure 9: Missing Stats

6.3.1 Remediation

If system is otherwise functional, this may not require immediate fixing, as Practable Portal can continue to function in this state, however the following steps can be attempted to try and remedy issue.

- Power Cycle Unit
- If you have access to the device CLI run:
 - `systemctl status practable-portal.service`
 - Ensure “enabled” is returned
- Contact Practable.io if situation persists

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7 Remote Rebooting

Although Practable Portals have proven to be highly reliable, with hundreds deployed across campus operating continuously, single-board computers are complex devices and may occasionally require a power cycle to restore normal operation.

If the device is locally accessible, this can be done by disconnecting and reconnecting the power supply. For installations where the device is located in a hard-to-reach area, the Practable Portal also supports remote power cycling, allowing it to power cycle itself and, where applicable, other connected devices.

7.1 Power Cycling Self

To power cycle a Practable Power unit, use the command line interface logged in as user practable.

1. Run command:

```
./practable-portal/safe_power_cycle.sh
```

2. Enter password for user practable
3. After ~3 seconds, the Info Screen will display *RX SBC Shutdown*. Power cycling procedure will commence after a pause to enable the system to safely shut down.

7.2 Power Cycling Neighbour Units

Power cycling neighbour units requires a two-wire interface (V+, GND) between each unit, from *shutdown-send* to *Vsbc-ext*.

We suggest wiring each unit to its neighbour in turn, and then wiring the last in the chain to the first in the chain. This way if a fault develops with any intermediate Portal that prevents access to the CLI, it can be externally power cycled by the unit before it in the chain. See Diagram below:

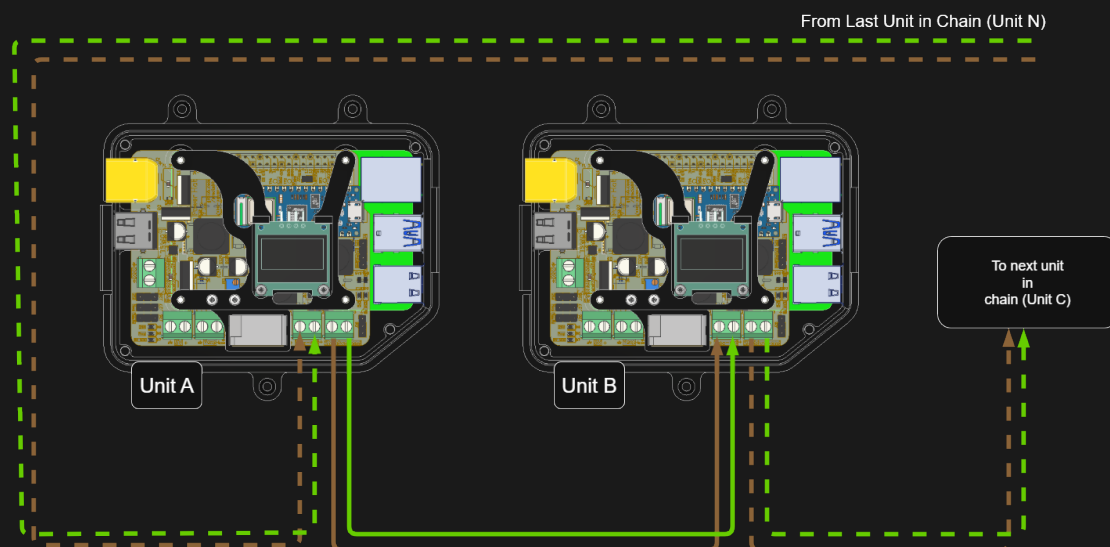


Figure 10: Recommended Wiring for Remote Rebooting

7.3 Power Cycling Neighbour Units

To power cycle a Practable Portal (Unit B) via its neighbour (Unit A), log into the command line interface of Unit A as user practable.

1. Run command:

```
./practable-portal/send_shutdown.sh
```

2. Enter password for user practable
3. After 3 seconds, screen of Unit B will display **RX SBC Shutdown**. Power cycling procedure will commence after a pause.

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